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Insect zoos and butterfly houses for public education: issues related to shipping and international trade of non-native insects

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Summary

Informal science education institutions such as zoos, natural history museums and botanical gardens exhibit live native and exotic insects and other arthropods to improve the general public's knowledge about these organisms and promote their conservation in nature. The purpose of this paper is to summarise the process of shipping exotic arthropods for exhibits and the regulations that apply, and to discuss issues that affect international shipment for this type of activity. These issues include escapes affecting the environment and delays affecting the viability of shipped insects. The regulatory agencies that issue permits for the importation of live insects for education and exhibit are discussed. The number of butterflies flying in the exhibits ranges from 100 specimens at the Butterfly Encounter of the Connecticut Science Center to a high of 15,000 specimens at the Dubai Butterfly Garden, with a mean of 2,048 specimens (n = 32). If the outlier of 15,000 is removed, the mean is 1,630 (n = 31). Insect zoos and butterfly exhibits play an overwhelmingly positive educational role by introducing millions of children and adults to the immensely important world of insects.

Keywords

Conservation – Education – Environmental impact – Importation – Insect displays – Public attitude – Regulation – Shipping – Wild collection.

Introduction

The purpose of this paper is to summarise the regulations and the process of importing and shipping exotic arthropods for exhibits such as insect zoos and butterfly houses and to discuss issues that affect international movement of live arthropods for this type of activity. Informal science education institutions such as zoos, natural history museums and botanical gardens exhibit live native and exotic insects and other arthropods to improve the general public's knowledge about these organisms and promote their conservation in nature [1, 2]. There are many institutions that focus on exhibiting native species of insects and arthropods found within the institution's geographical region, but these are not the focus of this paper.

Live exotic insects and other arthropods are obtained from commercial insect breeders, farmers or ranchers in tropical rainforests or other diverse ecosystems across the globe. They are shipped by these breeders via commercial airlines or international shippers such as FedEx to formal and informal science education exhibits around the world. These live animals are used for four primary purposes, namely to:

- improve public education
- support conservation programmes
- increase visits to an institution
- support research.

There are two general types of live insect exhibits:

- insect zoos or insectariums, which maintain individual species in separate glass or plexiglass enclosures of various sizes;
- butterfly exhibits or butterfly houses, which keep butterflies and moths in free-flight lepidopteries made of glass, plexiglass or, sometimes, screening material that the public can walk through.

Appendix 1 presents a range of such institutions from throughout the world, by region. This information was

collected through a Google search and is only representative of the range of operating facilities. (The COVID-19 pandemic has caused the temporary closure of some facilities at the time of writing.)

It can be difficult and expensive for institutions to rear certain insects, so there is a need for regular replenishment and introduction of new examples to provide continual displays. Collectors and producers may ship butterflies to individual users, but, more commonly, they ship batches of live insects to experienced aggregators in receiving countries who then redistribute smaller numbers domestically.

Maintaining live insect collections

Different institutions sustain their living exhibits in a range of ways. Insectariums such as Bugs Alive in Melbourne, Australia, the Montreal Insectarium in Canada, and the San Francisco Insect Zoo in the United States of America (USA) maintain their exhibit collection by choosing a combination of species with different life history traits, including:

- long-lived species, such as tenebrionid beetles, tarantulas and scorpions;
- social insect colonies such as leaf-cutter ants, honey bees and termites, which have long-lived queens and self-maintain large colonies if resources are provided;
- easy-to-breed species such as parthenogenic phasmids (walking sticks);
- other species that are also easy to breed, such as aquatic belostomatids (giant water bugs), diving beetles, mosquitoes, assassin bugs, burying beetles (*Nicrophorus* spp.) and tettigoniids (katydids);
- locally common native species that can be collected seasonally;
- other more challenging taxa, depending on the expertise of the staff.

Some colonies can be maintained for decades without inbreeding problems. Some exhibits, such as the Butterfly Conservatory in Niagara Falls, Canada, the Cockrell Butterfly Center in Houston in the USA, and the Butterfly Pavilion at the Artis Amsterdam Royal Zoo in the Netherlands breed a proportion of the species they exhibit on-site. However, most butterfly exhibits, whether they exhibit native or exotic species of butterflies and moths, have to purchase 100% of their pupae from suppliers, many of which have overseas sources.

Regulatory processes

All countries have import regulations that cover animals, which may be differentiated by taxonomic group and source country. For example, in the United Kingdom (UK), the Animal and Plant Health Agency issues licences for the importation of non-native insects (https://www.gov.uk/guidance/importing-non-native-animals). For imports from European Union (EU) countries, the rules covering insects are grouped along with the rules for reptiles, amphibians and other invertebrates, but treat honey bees and bumblebees separately from other insects (http://apha.defra.gov.uk/documents/bip/iin/bllv-9.pdf). The rules are directed at ensuring conservation of endangered species at source, preventing the importation of potential invasive species, and protecting plant and animal health. Pre-notification of shipments (between 1 and 30 days prior to planned arrival) is required and helps authorities to plan for inspection needs. At the port of entry (ports with appropriate facilities, such as London Heathrow Airport, are specified), they may be checked by government veterinarians before being released from inspection. Across the EU, regulations and processes are similar, except for a few countries.

In the USA, the United States Department of Agriculture (USDA) assesses permit applications for the importation of non-native insects. The assessment includes an evaluation of the species of insect in terms of its risk to the agriculture, floriculture and timber industries of the USA and to native ecosystems; the risk is assessed by considering the species' biology, diet breadth, dispersibility, reproductive potential and climatological range. The permit application assessment also includes an evaluation of the containment capabilities of the facility applying for the permit and of their 'Operation Protocol Manual', which describes the facility's operational procedures for maintaining the containment of all non-native insects listed on the permits. The Operation Protocol Manual contains information on personnel training, emergency procedures, the handling of waste material, and procedures that assist in preventing movement or escape of live non-native material, including pathogens, parasites and parasitoids. The facilities that maintain permits for non-native insects in the USA are generally inspected by a USDA representative annually.

The United States Fish and Wildlife Service (USFWS) has a list of 'designated ports' where international shipments can be received and inspected before they are permitted to enter. They are generally on either of the coasts or the border perimeters of the USA, with a few exceptions. The closest airports that would cover shipments from both Asia and Latin America are Los Angeles and San Francisco in California. Pupae from outside the USA are usually shipped as counter-to-counter air cargo, whereby the breeder in the exporting country takes the insects to the airport and the dealer expediter collects them when they arrive in the USA. Occasionally, insects are shipped by FedEx or UPS, but this is not preferred due to potential delays. The fee for USFWS inspection of a shipment of a species not covered by the Convention on International Trade in Endangered Species of Wild Fauna and Flora (CITES) was US$ 186 in 2014 [3] and, as of 2021, it is US$ 200 per shipment (http://www.lpsimports.com). This does not include

domestic shipping. Exhibit facilities in the USA must submit copies of their USDA Interstate Movement permits to customs authorities and the USDA along with invoices for the 'livestock' purchases. Currently, 90% of pupae ordered in the USA for butterfly exhibits are ordered directly through LPS LLC (http://www.lpsimports.com), a company that imports tropical pupae and insects for distribution across North America. It aggregates weekly orders from most butterfly exhibits in the USA, and some in Canada, into a break bulk pallet so that there is only one 'shipment item' to be inspected at the port and hence only one inspection fee from USFWS. After inspection and clearing customs, a broker then ships the pallet to the LPS facility in Denver, Colorado, where the individual orders are separated out and sorted into sub-shipments for the individual butterfly exhibit facilities. Any diseased and parasitised pupae are removed at this time. This process allows import shipments to be turned around quickly and they are shipped out via FedEx to arrive the next day if possible. This system allows exhibiters in the USA to order from multiple butterfly farms in the same shipment thanks to the coordination of shipping and port inspection. At the same time, shipping and inspection fees are shared. LPS handles all the necessary documents, such as the Declaration for Importation (https://www.fws.gov/media/wildlife-shipments-declaration-form-3-177).

Within the EU, movement of live insect shipments is more easily accomplished than it is across less connected borders, with the appropriate documents. Different cities in the EU can function as importation hubs for insect suppliers (primarily for pupae) from Latin America, Asia and Africa. These cities can receive aggregated bulk shipments that are then dispersed to individual butterfly exhibits. Stratford-upon-Avon, Amsterdam and Frankfurt are three such hubs. As well as having personnel to receive air freight and check regulatory documentation, these locations have the appropriate Veterinary Services to perform inspections to process the shipments efficiently.

The destination facilities may require licences and regulatory inspections in some countries. In the UK, for example, exhibit facilities are inspected every three years. There is a fee charged for this inspection (approximately £ 300) and for a pre-inspection audit. Some facilities are registered with a local council to operate as a zoo and so they receive a health and safety inspection along with the zoo licence check. United Kingdom facilities may export pupae to the EU and other countries; however, each country has its own set of document requirements.

Experience demonstrates that regular shipments from frequent suppliers through specialised aggregators at established ports of entry, with prior notification, can ensure relatively rapid transfer from exotic sources to the domestic recipient. Where recipients are recognised receivers of live insects, there may be further opportunities to streamline the process.

Conservation regulations

Shipments of live insects are governed by some specific conservation agreements and regulations to protect biodiversity in source areas. There is a suite of internationally recognised classifications that protect insects from over-collection. These include the International Union for Conservation of Nature (IUCN) Red List of Threatened Species [4], which shows species' conservation status, indicating if a species is endangered in its country of origin. There is a European equivalent list [5] and the IUCN Red List of Ecosystems [6], which analyses the risk of ecosystem collapse. Species and ecosystems under threat can be categorised as Critically Endangered, Endangered, Vulnerable or Near Threatened. In addition, species are assessed by changes in geographic distribution.

Before insects are exported, checks should also be made to ensure that the species is not listed under CITES. The species covered by CITES are listed in three appendices [7], according to the degree of protection they need.

- Appendix I includes species threatened with extinction. Trade in specimens of these species is permitted only in exceptional circumstances. Both import and export permits are required and will only be issued if the trade will not be detrimental to the survival of the species.
- Appendix II includes species not necessarily threatened with extinction, but in which trade must be controlled in order to avoid use that would be incompatible with their survival. An export permit is required, but an import permit is not.
- Appendix III contains species that are protected in at least one country, which has asked other CITES Parties for assistance in controlling the trade. If exported from the country in which the species is protected, an export permit is required. If exported from another country, a certificate of origin is required.

Trogonoptera brookiana and *Troides amphrysus* (Figure 1) are listed in CITES Appendix II, which states that any trade of the specimens must be controlled to avoid overexploitation of the resources [8]. As a result, similar but less endangered species are now used for education exhibits to promote public awareness and interest in conservation of the biodiversity of insects and of the wildlife ecosystems that support this biodiversity.

Country regulations have changed over the past several decades due to changes in species population numbers and changes in human attitudes towards insects. In Malaysia, which is one of the sources of live butterfly pupae for butterfly houses, the Wildlife Conservation Act 2010 lists

Idea lynceus *Troides amphrysus* *Trogonoptera brookiana*

Figure 1

Three of the butterfly species that are protected by the Malaysian Wildlife Conservation Act 2010

Highly valued by hobbyists and butterfly houses alike, these species are now protected by the Act and cannot be collected without a licence

Left to right: © Alex Dumchus; © Shutterstock Inc. Web. 9 January 2022; © Anaxibia (CC) CC BY-SA 3.0 (Filename: Trogonoptera brookiana 0588.JPG. Uploaded: 14 December 2016)

eight species as protected (*Trogonoptera brookiana*, *Troides amphrysus*, *Idea stolli*, *Zeuxidia aurelius*, *Agatasa calydonia*, *Charaxes durnfordi*, *Charaxes solon* and *Prothoe franck*). These species are vulnerable to illegal trade because of their great aesthetic appeal, and, before the Act was introduced, some were being harvested and exported globally [9]. With the enactment of the legislation, it is now illegal in Malaysia to harvest, keep or trade specimens of the protected species without a licence, even for research purposes [10]. The Malaysian giant tree nymph, *Idea lynceus*, is now also protected due to its large size and favoured status among hobbyists (Figure 1). In some facilities, *Idea leuconoe* is exhibited instead [2].

Scale of the industry

The number of insectariums and butterfly house exhibits changes over time [1] (Figure 2). The numbers in Figure 2 are based on a rapid online survey thus underestimate the actual number of butterfly houses and insect zoos in the UK, EU, and perhaps other areas of the world due to the author's survey bias and insufficient resources to mount a full correspondence–response survey. The highest proportion of regular (weekly) shipments of non-native insects are imported on behalf of butterfly house exhibits that are exhibiting exotic butterflies and moths in free-flight lepidopteries. Each facility orders between 250 and 400 pupae per week, depending on the size of the free-flight area (Appendix 1) and the financial resources of the facility. Some facilities are open year-round, and others are open seasonally. The number of butterflies flying in an exhibit ranges from around 100 to as many as 15,000 (at the Dubai Butterfly Garden), as stated on the websites of the different facilities. The mean from the survey results is 2,048 (n = 32). If the outlier of 15,000 is removed, the mean is 1,630 (n = 31).

Butterfly houses typically receive pupa shipments of lepidopteran species (butterflies and moths) weekly from Asia, Latin America and Africa. Rising transportation fees are increasingly becoming a barrier to operations. Attendance at public education institutions during the recent COVID-19 pandemic has been restricted, resulting in financial hardship both for the science education facilities that exhibit the live insects (e.g. museums, zoos, botanical gardens) and for the groups that supply them.

Risks posed by importation, and mitigation of risks

The various permitting or licensing agencies of each country assess the risk of each species application prior to approving importation. The risks from importation of non-native insects for exhibits are based on the possibility that the species imported may escape from the containment facility. The probability of survival and spread if there is an escape is based on biology, diet breadth, dispersibility, reproductive potential and predicted climate tolerance. The risk of importing a suite of pathogens (bacteria, viruses and protozoa), parasites and parasitoids is also evaluated, and this risk should be addressed and eliminated completely if possible. In addition, facilities are inspected routinely by their licensing agency.

Both insect zoos and butterfly houses mitigate risks by the design of their facilities, which are based on performance standards. Each facility may look different, but they must all 'perform' to contain the species that they exhibit. Examples of features that assist in containment are:

- double door entrances, or a combination of double doors, fans, lights and mirrors;
- fine screening on all air vents and sink drains;
- emergence cages with sleeves to contain parasites that may emerge from pupae;
- light traps to catch parasitoids in rearing areas.

In the author's experience, and from commentaries by other managers of insect facilities, the most important component

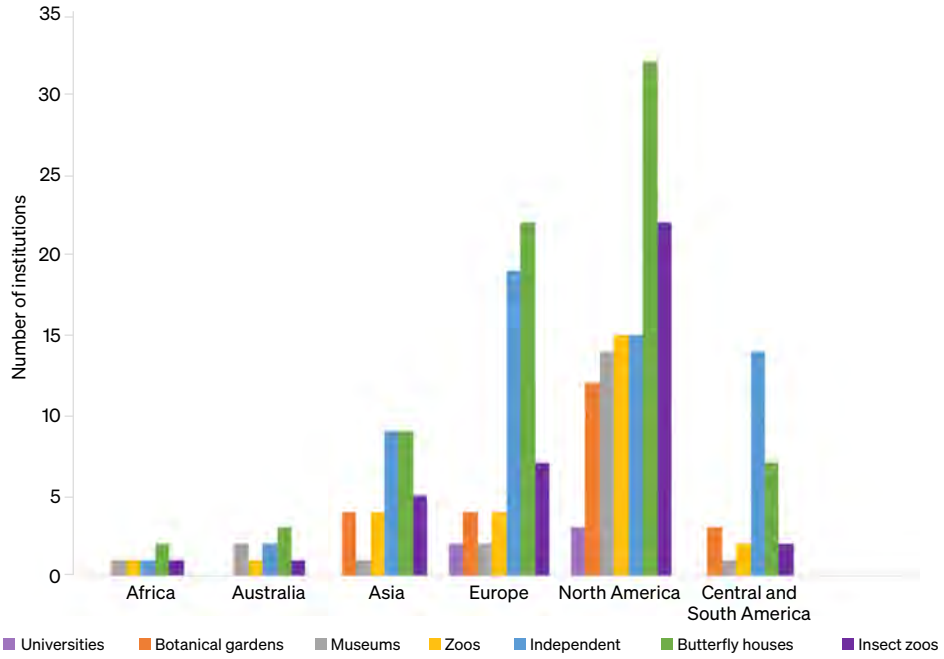

Figure 2

Numbers of insect exhibits around the world by geographical region and by institution type

Based on data summarised in the Appendix 1. Most of the exhibits in Australia, Central America and South America exhibit species from within their ecoregions

of facility performance is personnel training, the importance of which cannot be overstated. This ensures understanding of and commitment to the goals of containment and reduces human error. Personnel training and institutional accreditation for these institutions is available annually through various professional groups, such as the Terrestrial Invertebrate Taxon Advisory Group of the Association of Zoos and Aquariums, the International Conference of Butterfly Exhibitors and Suppliers, and the European Association of Zoos and Aquaria.

In some facilities all soil and waste material, including eggs and specimens, is incinerated, in other facilities it is double bagged and frozen, and in others it is autoclaved. Sterilising agents such as diluted solutions of sodium hypochlorite are used to clean equipment, rearing enclosures and working surfaces to continually manage the risk of pathogens.

Because the vast majority of organisms have restricted natural ranges due to the interaction of their ecology and increasingly fragmented ecosystems, the biological basis of the threat posed by the importation of exotic insects is the potential for their establishment in a new ecosystem that is free from their associated predators, parasites or parasitoids. If released from their historically limited ranges by translocation to new environments, exotic insects may find suitable climatic conditions and host plants that allow them to become established [11] and/or bring a pathogen or parasitoid with them if they escape containment. The key is to prevent the imported organisms from escaping from the containment facilities or during transport.

Importing country authorities may, therefore, decide to restrict the importation of certain species, based on their evaluation of whether or not the risk of escape and establishment of the non-native species, and associated parasite or pathogens, is significant. Licensing of approved recipients (with procedural requirements on containment, disposal, etc.) is a management option available in relation to such risks. Shipping to recognised licensed brokers may facilitate the movement of live insect shipments to the final institutional destination. Shipping directly to institutions with experience in import documentation and customs can also work, but may be more expensive than shipping with aggregators, as the latter can offer the benefits of sharing shipping costs and inspection fees.

Changing attitudes

Attitudes towards insects have changed immensely over the past two decades. Insects once feared in some parts of the world are now championed for their pollination services. Moreover, as populations of the ever-popular butterfly are in decline [12], there is now a greater focus on conservation. According to a 35-year dataset, even common species are falling in number [13]. While the public is unaware of the decline of most species, it is now familiar with the sharp decline of monarch butterfly populations in the western USA [14]. Due to this recent decline, government agencies have shifted regulations. Formerly, educators would breed and

release monarchs, but the California Department of Fish and Game has now made it illegal for anyone to collect a monarch butterfly for any purpose, even to breed and release it. Moving monarch butterfly individuals and populations to other locations represents a risk to native individuals and groups from genetic mixing and pathogens [11]. In some states, commercial breeders release some native species of butterflies for events such as weddings; however, some researchers believe this practice should end. Moore [15] states that individuals that originate non-locally will breed with local individuals and decrease the fitness of the local population by introducing genes that are not optimal for the local environmental conditions.

The USA allows breeding and release of nine species of native butterflies. However, only six species (*Vanessa cardui*, *Vanessa virginiensis*, *Nymphalis antiopa*, *Vanessa atalanta*, *Danaus plexippus* (west) and (east), and *Agraulis vanillae*), can be shipped and released in California [11]. The North American Butterfly Association (https://www.naba.org) does not support the movement and subsequent release into the environment of any commercially bred native species of butterflies for events (weddings, parties, etc.). In other countries, such as the UK, butterfly releases are also not allowed, due to similar concerns.

Conclusions

Despite changing attitudes towards collection and more interest in conservation of butterflies, interest in seeing live insects remains strong. Insect zoos and butterfly exhibits play an overwhelmingly positive educational role by introducing millions of children and adults to the immensely important world of insects each year. These insect exhibits range from large public institutions to small local venues. Many of the exhibits, across the range of scales, need to regularly replenish their stocks of exotic species, often on a weekly basis. This works smoothly where it is facilitated by aggregators in hubs that can then redistribute smaller lots to individual exhibits. Factors that contribute to the safe maintenance of exotic insects in exhibit facilities include species risk assessment, appropriate containment facilities, effective operational procedures and staff training. Regulations variously aimed at conservation, preventing the introduction of invasive species, and ensuring animal and plant health can differ considerably across different sources, destinations and species, making the shipping task more difficult and increasingly more expensive, especially for less experienced exporters and buyers.

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Insectariums et serres à papillons à visée éducative : questions liées au transport et au commerce international d'insectes exotiques

L. Saul-Gershenz

Résumé

Les institutions scientifiques à vocation éducative informelle telles que les parcs zoologiques, les muséums d'histoire naturelle et les jardins botaniques exposent nombre d'espèces tant natives qu'exotiques d'insectes et d'autres arthropodes afin de mieux les faire connaître du grand public et de promouvoir leur conservation dans la nature. Cet article a pour objet de présenter succinctement les modalités de l'expédition d'arthropodes exotiques destinés aux espaces d'exposition ainsi que les réglementations applicables, et d'examiner les problèmes affectant les expéditions internationales d'insectes dans ce type d'activités. Les problèmes relevés concernant principalement les évasions d'insectes et leurs effets sur l'environnement, et les retards de livraison et leurs effets sur la survie des insectes transportés. L'autrice évoque également les agences de réglementation chargées de délivrer les autorisations d'importer des insectes à des fins éducatives et d'exposition. Le nombre de papillons vivant dans des serres d'exposition va des 100 spécimens du site du Connecticut Science Center aux 15 000 spécimens du Jardin aux papillons de Dubaï, la moyenne étant de 2 048 spécimens par site (n = 32). Si l'on retire le cas particulier du site de Dubaï, la moyenne par

site est de 1 630 spécimens (*n* = 31). Les insectariums et les serres à papillons jouent un rôle éducatif essentiel en permettant à des millions d'enfants et d'adultes de découvrir le monde des insectes et leur rôle fondamental.

Mots-clés

Attitude du public – Collections sauvages – Conservation – Éducation – Impact environnemental – Importation – Présentations d'insectes – Réglementation – Transport.

Insectarios y mariposarios con fines de educación popular: cuestiones ligadas al transporte y el comercio internacional de insectos alóctonos

L. Saul-Gershenz

Resumen

Los establecimientos científicos que ejercen una labor pedagógica oficiosa, como zoológicos, museos de historia natural o jardines botánicos, exponen especies autóctonas y alóctonas de insectos y otros artrópodos para familiarizar al gran público con ellos y promover su conservación en el medio natural. La autora, tras presentar sucintamente el proceso de expedición de artrópodos exóticos con destino a espacios de exposición y los reglamentos aplicables, examina una serie de cuestiones que inciden en este tipo de transporte internacional, principalmente las fugas de ejemplares y sus consecuencias ambientales y los retrasos que afectan a la supervivencia de los insectos transportados. También se detiene en los organismos de reglamentación que expiden los permisos de importación de insectos vivos con fines pedagógicos o de exposición. El número de mariposas expuestas puede ir desde los 100 ejemplares presentados en el Connecticut Science Center hasta los 15 000 que alberga el mariposario de Dubai, con un promedio que se sitúa en los 2 048 ejemplares (*n* = 32), aunque, si no se tiene en cuenta el valor atípico de Dubai, la media resultante es de 1 630 ejemplares (*n* = 31). Los insectarios y mariposarios cumplen una extraordinaria función pedagógica, pues millones de niños y adultos descubren en ellos el importantísimo mundo de los insectos.

Palabras clave

Actitud del público – Captura salvaje – Conservación – Educación – Exposiciones de insectos – Impacto ambiental – Importación – Reglamento – Transporte.

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Appendix 1
Insect zoos, butterfly house exhibits and suppliers around the world

Information was obtained in 2021 using the Google search engine. This is not an exhaustive list but one that represents the diversity of types of shippers and receivers. Data provided is as presented on the websites. Where information is incomplete, it was not available online. Websites were accessed and available in January 2022.

Abbreviations:

BB:	butterfly breeding	GOV:	government	NP:	national park
BE:	butterfly exhibit	I:	independent	NR:	nature reserve
BF:	butterfly farm	IZ:	insect zoo or insectarium	S:	seasonal operation
BG:	botanical garden	M:	within a museum	U:	university
BH:	butterfly house	N:	native	Y:	year-round operation
E:	exotic	NC:	nature centre	Z:	within a zoo

	Institution	Website or online information	No. of butterfly specimens/No. of butterfly species	Institution type	Native and/or exotic spp.	Year opened/ size/ notes
Australia & New Zealand						
1	Australian Butterfly Sanctuary Kuranda, Australia	www.kuranda.org/listing/australian-butterfly-sanctuary		I, BH	N	
2	Coffs Harbour Butterfly House Bonville, Australia	www.butterflyhouse.com.au		I, BH , Y		
3	Bugs Alive, Museums Victoria, Melbourne Museum Carlton, Australia	https://museumsvictoria.com.au/melbournemuseum/whats-on/bugs-alive		M, IZ		
4	Butterfly House, Melbourne Zoo Parkville, Australia	www.zoo.org.au/melbourne/whats-on/news/melbourne-zoos-iconic-butterfly-house-reopens	600/13	Z, BH		1985
5	Otago Museum Tūhura Tropical Forest Dunedin, New Zealand	https://otagomuseum.nz/whats-on/tuhura/tropical-forest		M		
Africa						
1	Algiers Insectarium Algiers, Algeria	https://maliactu.net/linsectarium-dalger-a-la-pointe-de-la-lutte-contre-les-ravageurs/?__cf_chl_managed_tk__=pmd_yfkiDNEt.NJi6._y0elggeijX9tmXqbXpUoz6B6e5pA-1632295966-0-gqNtZGzNAzujcnBszRQI		M, IZ		1924 Insect museum and biocontrol research
2	Exotic Animal World (formerly Butterfly World Zoo) Stellenbosch, South Africa	https://exoticanimalworld.co.za		Z, BH		1996 International importation of butterflies stopped due to COVID-19 pandemic
3	Kipepeo Butterfly Project & Mombasa Butterfly House Malindi, Kenya	www.kipepeo.org	/17	IZ, BH, BB	N	
4	Amani Butterfly Project Amani, Tanzania	www.minibeast.uk/amani		BB		Supplier
5	Mida Butterfly Farm Kenya Watamu, Kenya	www.naturalhistorycuriosities.com/conservation/supporting-the-mida-butterfly-farm-in-watamu-kenya	/50-70	BB		1990

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Asia						
1	Banteay Srey Butterfly Centre Siem Reap, Cambodia	https://en.wikipedia.org/wiki/Banteay_Srey_Butterfly_Centre		I, BH, BB	N	2009
2	Butterfly Conservatory of Goa Goa, India	www.ifoundbutterflies.org/butterfly-conservatory-goa		I, BH		400 m², 1,000 m²
3	Butterfly Park, Bannerghatta National Park Bangalore, India	https://bannerghattabiologicalpark.org/butterfly_park.html		NP	N	1,000 m²
4	Butterfly Garden, Kadoorie Farm and Botanic Garden, Hong Kong, Special Administrative Region of the People's Republic of China	www.kfbg.org/en/attractions/Butterfly-Garden		BG		
5	Butterfly House, Ocean Park Hong Kong, Special Administrative Region of the People's Republic of China	www.chinaculturetour.com/hongkong/ocean-park.htm		I, BH		
6	Butterfly Park and Insect Kingdom Sentosa, Singapore	www.hotels.com/go/singapore/singapore-butterfly-kingdom	1,500/50	Z, IZ		
7	Dubai Butterfly Garden, Dubai Miracle Garden Dubai, United Arab Emirates	www.dubaibutterflygarden.com	15,000/50	I, BH	E	6,673 m²
8	Entopia, Penang Butterfly Farm Pulau Pinang, Malaysia	www.entopia.com	5,000/120	I, BB, BH, Y		
9	Gunma Insect World Kiryu, Japan	www.giw.pref.gunma.jp www.kiea.jp/GunmaInsectWorld.html		I, NR, Y		2005 4,309 m²
10	Itami City Museum of Insects Itami, Japan	www.itakon.com		M		
11	Insectarium, Butterfly Hot House, Tama Zoological Park Tokyo, Japan	https://japan.apike.ca/japan_tokyo_tama_zoo.html		Z, IZ		1,104 m²
12	Insectarium, Taipei Zoo Taipei City, Chinese Taipei	https://english.zoo.gov.taipei/News_Content.aspx?n=C9FB393E85F4A2B6&sms=DFFA119D1FD5602C&s=4AB593B171CFB866	/400	Z, IZ	N & E	
13	Konya Tropical Butterfly Garden Konya, Turkey	https://www.konyatropikalkelebekbahcesi.com/en	/15	BG, BE		1,600 m²
14	Phuket Butterfly Garden & Insect World Phuket, Thailand	https://phuketindex.com/butterflygarden		I, IZ, BG		
15	Melaka Butterfly and Reptile Sanctuary Ayer Keroh, Malaysia	www.butterflyreptile.com	/20	I, BG, BE		1991
16	Simply Butterflies Conservation Center Bilar, Philippines	www.boholtourismph.com/simply-butterflies-conservation-center		I, BH		
17	The Fragile Forest Enclosure, Singapore Zoological Gardens Singapore	www.mandai.com/en/singapore-zoo/animals-and-zones/fragile-forest.html		Z, IZ, BE		
18	Tropical Butterfly Conservatory Tiruchirappalli, India	https://en.wikipedia.org/wiki/Tropical_butterfly_conservatory,_Trichy		I, BH		

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North America & Mexico						
1	Aveda Butterfly Garden, Minnesota Zoo, Apple Valley, United States of America	https://mnzoo.org/blog/tag/aveda-butterfly-garden	/40	Z, BE, S	N	
2	Bayer Insectarium & Mary Ann Lee Butterfly Wing, Saint Louis Zoological Park Saint Louis, United States of America	www.stlzoo.org/visit/thingstoseeanddo/discoverycorner/insectarium		Z, BH, IZ		2000
3	Butterfly Garden, Bronx Zoo New York, United States of America	https://bronxzoo.com/things-to-do/exhibits/butterfly-garden		Z, BE, S	N	
4	Butterfly exhibit, American Museum of Natural History New York, United States of America	www.amnh.org/exhibitions/butterflies	500/	M, BE, S	E	
5	Butterfly House, Children's Museum of Southern Minnesota Mankato, United States of America	www.cmsouthernmn.org/butterfly-house		M, BE	N	
6	Audubon Butterfly Garden and Insectarium New Orleans, United States of America	https://audubonnatureinstitute.org/insectarium		Z, IZ		
7	Bear Mountain Butterfly Sanctuary Jim Thorpe, United States of America	https://bearmountainbutterflies.com/raising-butterflies		I, BE	N	Painted lady and monarch butterfly releases
8	Berniece Grewcock Butterfly and Insect Pavilion, Henry Doorly Zoo and Aquarium Omaha, United States of America	www.omahazoo.com/berniece-grewcock-butterfly-and-insect-pavilion		Z, IZ, BH	E	
9	Butterfly House and Insect World Mackinac Island, United States of America	www.originalbutterflyhouse.com	500-700/	I, BH, S		1991 167 m^2
10	Blooming Butterfly Garden, Como Zoo & Conservatory Saint Paul, United States of America	https://www.youtube.com/watch?v=QJDtYGcL7es	1,000/100	Z, BE	E	232 m^2
11	Bug House, Houston Zoo Houston, United States of America	www.houstonzoo.org/explore/exhibits/bug-house		Z, IZ		2014
12	Butterfly Biosphere, Thanksgiving Point Lehi, United States of America	https://thanksgivingpoint.org/experience/butterfly-biosphere	1,000/	I, BH	E	3,716 m^2
13	Butterfly Conservatory & Insect Zoo, Kansas State University Manhattan, United States of America	www.k-state.edu/gardens/education/insect-zoo	100-200/20-30	U, IZ		
14	Butterfly Encounter, Connecticut Science Center Hartford, United States of America	https://ctsciencecenter.org/butterfly-encounter		M, BE	E	
15	Butterfly House at Churchville Nature Center Churchville, United States of America	www.localgymsandfitness.com/US/Southampton/499910143494954/The-Butterfly-House-at-Churchville-Nature-Center		NC, BH, IZ, S	N	2014
16	Butterfly Garden & Insect Zoo, Museum of Science Boston, United States of America	www.mos.org/explore/exhibits/butterfly-garden		M, IZ		
17	Butterfly Habitat, Six Flags Discovery Kingdom Vallejo, United States of America	www.sixflags.com/discoverykingdom/attractions/butterfly-habitat		I, BH	E	
18	Butterfly House, Ashland Nature Center Hockessin, United States of America	www.onlyinyourstate.com/delaware/ashland-nature-center-de		NC, BH	N	

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19	Butterfly House, Detroit Zoo Royal Oak, United States of America	https://detroitzoo.org/animals/zoo-animals/butterfly	250 pupae/ week/25	Z, BH	E	
20	Butterfly House, San Antonio Zoo San Antonio, United States of America	www.tripadvisor.com/ ShowUserReviews-g60956-d104199-r472224489-San_Antonio_Zoo-San_Antonio_Texas.html	/15-20	Z, BH		156 m^2
21	Butterfly House, Wheeler Farms Whitehouse, United States of America	www.wheelerfarms.com/butterfly-house		I, BH	E	
22	Butterfly House, The Gardens on Spring Creek Fort Collins, United States of America	www.fcgov.com/gardens	100/	I, BH, Y	N	2019 158 m^2
23	Butterfly Hollow, Franklin Park Zoo Boston, United States of America	www.zoonewengland.org/franklin-park-zoo/exhibits/butterfly-hollow-seasonal	/40	Z, BE, S	N	
24	Butterfly Magic, Tucson Botanical Gardens Tucson, United States of America	https://tucsonbotanical.org		BG, S	E	
25	Butterfly Palace and Rainforest Adventure Branson, United States of America	www.thebutterflypalace.com	1,000/40-60	I, BH, Y		650 m^2 Spends US$ 70-100,000 per year on pupa
26	Butterfly Pavilion Westminster, United States of America	https://butterflies.org	1,200/	I, BH, IZ	E	1995 9,144 m^2; Wings of the Tropics, 650 m^2
27	Butterfly Place Westford, United States of America	https://butterflyplace-ma.com		I, BH	E	
28	Butterfly Jungle, San Diego Wild Animal Park Escondido, United States of America	https://sdzsafaripark.org/animals/butterfly		Z, BH, S	E	
29	Butterfly Rainforest, Florida Museum of Natural History Gainesville, United States of America	www.floridamuseum.ufl.edu/exhibits/butterfly-rainforest	1,000/50	M, BH	E	595 m^2
30	Butterfly Wing, Reiman Gardens, Iowa State University Ames, United States of America	www.reimangardens.com/education-ideas/butterfly-insect-links-and-articles	800/80	U, BE	N & E	232 m^2
31	Butterfly Wonderland Scottsdale, United States of America	https://butterflywonderland.com	100/	I, BE	E	
32	Cambridge Butterfly Conservatory Cambridge, Canada	www.cambridgebutterfly.com	2,000/57	BH, IZ	E	2001 In partnership with El Bosque Nuevo Butterfly Farm, 100% of the profits from the purchase of butterflies is invested into reforestation and conservation of tropical rainforest habitat in the Guanacaste region of Costa Rica
33	Cecil B. Day Butterfly Center, Callaway Resort and Gardens Pine Mountain, United States of America	www.callawaygardens.com/the-gardens/places-to-explore/day-butterfly-center	1,000/	BG, BH, Y	E	
34	Cockrell Butterfly Center, Houston Museum of Natural Science Houston, United States of America	www.hmns.org/cockrell-butterfly-center	/11	M, BH, Y	E	

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35	Frederik Meijer Gardens & Sculpture Park, Fred & Dorothy Fichter Butterflies are Blooming Exhibition Grand Rapids, United States of America	www.meijergardens.org/calendar/butterflies-are-blooming	7,000/60	BG, BH, S	E	Lena Meijer Conservatory, 1,400 m²
36	Insect Zoo & Butterfly Pavilion, Natural History Museum of Los Angeles County Los Angeles, United States of America	https://nhm.org/experience-nhm/exhibitions-natural-history-museum/butterfly-pavilion		M, IZ, Y, BH, S	N	
37	Insect House, San Diego Zoo San Diego, United States of America	www.shyguysworld.com/index.php?topic=20802.0		Z, IZ	E	
38	Idlewild Butterfly Farm and Insectarium Louisville, United States of America	www.idlewildbutterflyfarm.com/butterflies		BF, IZ, BH	N	2009 Releases butterflies at events
39	Insect Zoo, San Francisco Zoo San Francisco, United States of America	www.sfzoo.org/australian-walking-stick	1,000/20	Z, IZ, Y	N & E	1979
40	Insect Exhibit, Toronto Zoo Toronto, Canada	www.torontozoo.com/welcomeback		Z, IZ		Insect exhibits within other exhibits
41	Key West Butterfly and Nature Conservatory Key West, United States of America	www.keywestbutterfly.com	/50-60	I, BH	E	
42	Living Conservatory, North Carolina Museum of Natural Sciences Raleigh, United States of America	https://naturalsciences.org/living-collections/living-conservatory		M, IZ		
43	Magic Wings Butterfly Conservatory & Gardens South Deerfield, United States of America	https://magicwings.com	3,000-4,000/	BG, BH	E	2000 743 m²
44	Magic Wings Butterfly House & Bayer Insectarium, North Carolina Museum of Life & Science Durham, United States of America	https://www.lifeandscience.org/explore/butterfly-house	1,000/30-50	M, BH, IZ, Y	E	
45	Marshall Butterfly Pavilion, Desert Botanical Garden Phoenix, United States of America	https://dbg.org/exhibits/mighty-monarchs/2021-09-25		BG, BE		
46	Michigan State University, Horticultural Gardens & Butterfly House East Lansing, United States of America	www.michigan.org/property/michigan-state-university-horticultural-gardens-and-butterfly-house		U, BG, BH, S		
47	Montreal Insectarium Montreal, Canada	https://espacepourlavie.ca/en/about-insectarium		I, IZ, Y		
48	Newfoundland Insectarium Reidville, Canada	https://nlinsectarium.com	100/	M, IZ, BH, S		1998
49	Niagara Park Butterfly Conservatory Niagara, Canada	www.niagaraparks.com/visit/attractions/butterfly-conservatory	2,000/45	I, BG, BH GOV	E	180 m²
50	Orange County Native Butterfly House, The Environmental Nature Center Newport Beach, United States of America	https://encenter.org/visit-us/butterfly-house		NC, BH, S	N	
51	Osher Rainforest, California Academy of Sciences San Francisco, United States of America	www.calacademy.org/exhibits/osher-rainforest	1,250/	M, BE, IZ	E	

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52	O. Orkin Insect Zoo & Butterfly Pavilion, Smithsonian Institution, National Museum of Natural History Washington, DC, United States of America	www.si.edu/exhibitions/insect-zoo-o-orkin%3Aevent-exhib-132	/16	M, IZ, Y, BE	N & E	1976
53	Panhandle Butterfly House Milton, United States of America	https://panhandlebutterflyhouse.org		NC, BG	N	2021
54	Butterfly House, Paul Smith's College Visitor Interpretive Center, Adirondacks Park Paul Smiths, United States of America	www.adirondackvic.org/Butterfly-House.html		I, BH, S	N	1993
55	Puelicher Butterfly Wing, Milwaukee Public Museum Milwaukee, United States of America	www.mpm.edu/index.php/exhibitions/permanent-exhibits/first-floor-exhibits/puelicher-butterfly-wing	1,000/60	M, BH	N & E	
56	Sophia M. Sachs Butterfly House, Missouri Botanical Garden Chesterfield, United States of America	www.missouribotanicalgarden.org/visit/family-of-attractions/butterfly-house/about-the-butterfly-house.aspx	1,500/60-80	BG, BH	E	1998 743 m^2
57	Sertoma Butterfly House Sioux Falls, United States of America	https://butterflyhouseaquarium.org	800/	I, BH	E	334 m^2
58	Butterfly World & Bug Zoo, Tradewinds Park Coconut Creek, United States of America	www.butterflyworld.com/our-story	~5,000/80	I, BH, IZ	E	1988
59	Tropical Butterfly House, Pacific Science Center Seattle, United States of America	www.pacificsciencecenter.org/exhibits/butterfly-house		M, BE	E	
60	Western Colorado Botanical Gardens & Butterfly House Grand Junction, United States of America	https://wcbotanic.org/butterflies		BG, BH		1998
61	Victoria Bug Zoo Victoria, Canada	www.victoriabugzoo.ca		I, IZ		
62	Victoria Butterfly Gardens Brentwood Bay, Canada	www.butterflygardens.com	400-700/week (~4,000 total)/75	BG, BH	E	1,100 m^2
63	Wings of Fancy, Butterfly and Caterpillar Exhibit Brookside Gardens Wheaton, United States of America	https://montgomeryparks.org/parks-and-trails/brookside-gardens/wings-fancy-live-butterfly-caterpillar-exhibit		P, BE, S	N	
64	World of the Insect, Cincinnati Zoo Cincinnati, United States of America	https://cincinnatizoo.org/plan-your-visit/exhibits/world-of-the-insect		Z, IZ		1978
65	Fairchild Tropical Botanic Garden, Clinton Family Conservatory Miami, United States of America	https://fairchildgarden.org/miami-botanical-garden/butterfly-garden-miami	100/40	BG, BH	E	
66	Butterfly Dan's Kissimmee, United States of America	www.butterflydans.com		BB	N	Supplier
67	Zoológico Miguél Álvarez del Toro (formerly Zoológico de Tuxtla Gutiérrez) Chiapas, Mexico	https://en.wikipedia.org/wiki/Zool%C3%B3gico_Migu%C3%A9l_%C3%81lvarez_del_Toro		Z, IZ	N	

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Central & South America, Caribbean						
1	Butterfly Farm Orangestad, Aruba	www.thebutterflyfarm.com		I, BH, Y	E	
2	Cali Zoo, Butterfly House Cali, Colombia	www.zoologicodecali.com.co		Z, BH		450 m²
3	Chaa Creek Butterfly Farm San Ignacio, Belize	www.chaacreek.com/amenities/blue-morpho-butterfly-farm		I, BG, BH, BB		Supplier
4	Costa Rica Entomological Supply Guácima, Costa Rica	www.butterflyfarm.co.cr	/49	I, BB, BH		1984 Supplier
5	El Bosque Nuevo San José, Costa Rica	www.elbosquenuevo.org/copy-of-purchase-pupaes	/87	I, BB		Supplier
6	El Mariposario, Insectarium Calarcá, Colombia	https://jardinbotanicoquindio-org.translate.goog/?_x_tr_sl=es&_x_tr_tl=en&_x_tr_hl=en&_x_tr_pto=sc		I, BG, IZ, BB		640 m²
7	Green Hills Butterfly Ranch Mountain Pine Ridge Reserve, Belize	www.greenhillsbelize.com	/20	I, BG, BB	N	Supplier
8	Heliconius Butterfly Works Quito, Ecuador	www.heliconiusworks.com		I, BB, BF	E, N	1996 Supplier
9	Mariposario, Chapultepec Zoo Mexico City, Mexico	www.atlasobscura.com/places/mariposario-chapultepec http://data.sedema.cdmx.gob.mx/zoo_chapultepec	1,000/	I, BB		
10	Mariposas de Mindo Mindo, Ecuador	www.mariposasdemindo.com/index-en.html		I, BB		
11	Mariposario Tambopata Butterfly Farm Puerto Maldonado, Peru	www.livinginperu.com/puerto-maldonados-butterfly-farm	/15-30	I, BF, BB, BH		600 m²
12	Morpho Pupae/Mroch Neotropical S.R.L. San Ramón, Costa Rica	www.morphopupae.com		I, BB		Supplier
13	Pilpintuwasi Butterfly Farm Iquitos, Peru	https://pilpintuwasi.com		I, BF, BB		
14	Borbolária de São Paulo São Paulo, Brazil	https://eco.cidadedesaopaulo.com/atracoes/borboletario	/17	I, BH	N	
15	Spirogyra Jardín de Mariposas San José, Costa Rica	www.butterflygardencostarica.com	/37	I, BH, B		1992 7,000 m²
Europe						
1	Benalmadena Butterfly Park Málaga, Spain	www.mariposariodebenalmadena.com/en/mariposas	1,500/150	I, BH	E	
2	Bornholm Butterfly Park, Bornholms Sommerfuglepark Bomholm, Denmark	www.sommerfugleparken.dk	1,000/15-30	I, BG, S	E	1997 900 m²
3	Botanical Garden of P. J. Šafárik University Košický Kraj, Slovakia	www.upjs.sk/pracoviska/botanicka-zahrada	1,000-3,000/42-89	U, BH, BG, S		310 m²
4	Butterfly Arc, Esapolis Insectarium Padua, Italy	www.micromegamondo.com/en/museums-and-parks/esapolis		I, IZ	E	
5	Butterfly Botania, University of Eastern Finland Joensuu, Finland	www.botania.fi		U, BG	E	
6	Butterfly Dome Druskininkai, Lithuania	www.facebook.com/Butterfly-Dome-Drugeli%C5%B3-kupolas-1751829641511885		I, BH	E	
7	Butterfly House Praid, Romania	www.butterflyhouse.ro		I, BH	E	

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8	Butterfly Paradise & Bugs, London Zoo London, United Kingdom	www.zsl.org/zsl-london-zoo/exhibits/butterfly-paradise		Z, IZ, BH, Y	E	2006
9	Butterfly Park Empuriabrava Girona, Spain	www.butterflypark.es		I, BH	E	
10	Butterfly Pavilion & Insectarium, Artis Royal Amsterdam Zoo Amsterdam, the Netherlands	http://artis.nl		Z, IZ, BH, Y	E	
11	Butterfly World, Preston Park Stockton-on-Tees, United Kingdom	https://en.wikipedia.org/wiki/Preston_Park,_Stockton-on-Tees#	/80-120	I, BH		
12	Edinburgh Butterfly and Insect World Edinburgh, United Kingdom	www.edinburgh-butterfly-world.co.uk		I, BH, IZ	E	1984 Closed permanently in 2021 due to COVID-19 pandemic
13	Garden of the Butterflies, Sayn Castle Bendorf, Germany	www.sayn.de/garten-der-schmetterlinge		I, BH	E	1987
14	Garden of Butterflies Friedrichsruh Aumühle-Friedrichsruh, Germany	www.gartenderschmetterlinge.de	1,000/40	I, BH	E	1985 450 m^2
15	Insectarium & Butterfly Hall, Wilhelma Zoo Stuttgart, Germany	www.wilhelma.de/en/animals-and plants/animals/insectarium.html		Z, IZ	E	1979 IZ
16	Jardin des Papillons Grevenmacher, Luxembourg	www.papillons.lu	100/16	BG, BH		600 m^2
17	La Serre aux Papillons La Queue-lez-Yvelines, France	www.serreauxpapillons.com/pages/accueil	700/	I, BH, S	E	1989
18	Maximilianpark Hamm, Germany	www.maximilianpark.de/en/aufsichtsrat	/80	I, BH	E	450 m^2
19	Micropolis, La Cité des Insectes Saint-Léons, France	www.micropolis-aveyron.com		I, IZ		
20	Musée d'Histoire naturelle et Vivarium de l'Hôtel de ville Tournai, Belgium	https://mhn.tournai.be/collections		M, IZ	E	Live arthropods
21	Papiliorama Kerzers, Switzerland	www.papiliorama.ch/en	1,000/120 & 10N	I, BH, Y	E & N	1989 1,200 m^2; Bug Bigtop, 500 m^2 Papiliorama Foundation conservation programme also protects 400 km^2 of tropical nature in the Shipstern Nature Reserve in Belize
22	Papilonia Butterfly House Prague, Czech Republic	https://papilonia.cz/en	600/	I, BH	E	150 m^2
23	Passiflorahoeve Harskamp, the Netherlands	www.passiflorahoeve.nl		I, BF, S		1,600 m^2
24	Schmetterlinghaus Wien, Austria	www.schmetterlinghaus.at	500/40	I, BH		1990
25	Stratford Butterfly Farm Stratford-upon-Avon, United Kingdom	www.butterflyfarm.co.uk/attraction	2,000/250	I, BH	E	Pupae aggregator
26	Vlinders aan de Vliet Leidschendam, the Netherlands	www.vlindersaandevliet.nl		S, BH, S	E	1,600 m^2
27	Vlindersafari Vinderkas Rotterdam, the Netherlands	www.vlindersafari.nl		I, BH	E	
28	Wildlands Adventure Zoo Emmen, the Netherlands	www.wildlands.nl		Z, BH		